

NOISE OUT, BIAS IN: TARGETED BIAS INJECTION IN DIFFUSION LANGUAGE MODELS VIA CLOSED-LOOP ACTIVATION STEERING

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ABSTRACT

Masked diffusion language models (dLLMs) generate text by iteratively denoising masked positions, re-predicting each token multiple times before it is committed. An autoregressive decoder exposes an answer’s distribution once, at the step that commits it; a dLLM exposes it at every denoising step before commitment, and we show that an adversary can exploit this. Since an answer remains open to revision over many denoising steps, an adversary with access to internal activations can watch how likely the model is to produce a chosen answer and adjust the intervention accordingly. Building on this observation, we study *targeted bias injection*, an attack that steers a frozen dLLM toward a demographic answer selected by the adversary. The attack uses a simple proportional-integral (PI) controller that tracks the target-answer probability during denoising and adapts the strength of a steering vector on the fly. On ambiguous BBQ questions where the correct answer is abstention, our attack raises LLaDA-8B-Instruct’s preference for the targeted group from 1.8 to 16.7 percentage points, more than three times the strongest fixed-strength steering baseline, and on SocialStigmaQA it raises the selection of stigmatizing answers from 17.6% to 58.1%. Fitted to other demographic targets, the same attack shifts answers by up to 37 percentage points, and each attack takes about 40 minutes on one GPU. On the primary target, feedback is what makes the attack work: constant steering at the same average strength over the token-committing steps produces a far smaller shift while corrupting nearly three times as many outputs, and a constant strength set separately for each example still falls well short. Our findings identify the denoising trajectory as a new control channel in dLLMs and call for bias audits that examine the serving stack rather than the frozen model alone. Our code is publicly available on GitHub.

Warning: This paper contains examples of stereotyped and stigmatizing content about demographic groups.

1 INTRODUCTION

Diffusion large language models (dLLMs) generate text by iteratively denoising a masked sequence instead of predicting one token after another (Li et al., 2022; Austin et al., 2021; Lou et al., 2024; Sahoo et al., 2024). Open models such as LLaDA (Nie et al., 2025), Dream (Ye et al., 2025), and LLaDA-MoE (Zhu et al., 2025) now match autoregressive models of similar scale, which makes them candidates for the same uses, from customer-facing assistants (Brynjolfsson et al., 2025) to hiring and other screening decisions (Burema et al., 2026). Because they are trained on similar corpora and instruction-tuned in similar ways (Ouyang et al., 2022), there is no reason to expect them to be any fairer. Autoregressive LLMs carry stereotypes from their training data into their answers (Thaler et al., 2026; Nadeem et al., 2021; Li et al., 2020; Parrish et al., 2022), in free generation (Wan & Chang, 2025; Pan et al., 2025; Wang et al., 2026) as much as in question answering (Jin et al., 2025) and across languages (Mitchell et al., 2025; Saeed et al., 2026; Gamboa et al., 2025), and alignment

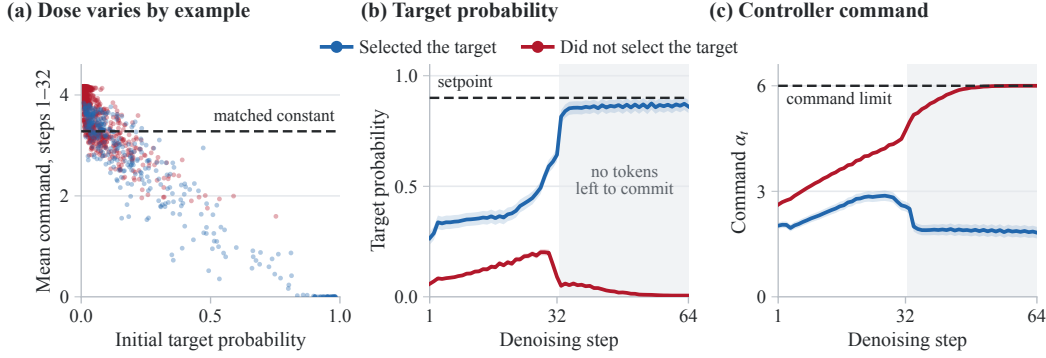


Figure 1: **Feedback assigns each example its own steering strength.** Primary Black-target PI run on LLaDA-8B-Instruct, 1,200 sequences, split by strict outcome. (a) Each point is one sequence: its target probability after the first controlled step against its mean command over steps 1–32; the dashed line is the effort-matched constant. (b,c) Mean target probability and command per step with 95% bands. Shading marks steps after the last token commits.

removes explicit bias more reliably than the implicit associations behind it (Bai et al., 2025; Zhao et al., 2025b). The same pattern has already been seen in dLLMs. On an income-prediction task, TrustLDM (Mo et al., 2026) finds gender disparities in both LLaDA and Dream, and a malicious context makes them worse.

Audits of this kind measure the bias that training leaves in a model’s weights. Bias can also be added later, at inference time, by whoever controls the model’s inputs or its serving stack. For autoregressive LLMs, crafted prompts can pull answers toward a chosen demographic group (Zhao et al., 2025a), and so can a steering vector added to the residual stream (Wang & Shu, 2024). The second route goes through the same forward hooks that activation steering (Turner et al., 2023; Rinsky et al., 2024; Li et al., 2023; Zou et al., 2023; Ardit et al., 2024; Rodríguez et al., 2025; Suau et al., 2024) uses for benign control and FairSteer (Li et al., 2025) uses for debiasing, and it leaves the weights and the user’s prompt exactly as they were, so an audit of the checkpoint or of the prompt pipeline finds nothing. Whether a deployed system is biased therefore depends on its serving stack as well as its checkpoint, and for dLLMs that stack has not been examined at all.

Examining it is not a matter of directly using the autoregressive attack, because the two decoders give an attacker different information. Nguyen et al. (2026) have shown that activation steering is a control problem. Here the output is the probability of the target answer at the unresolved position, and the input is the steering strength α . An autoregressive model exposes that probability only when it commits the token, so α has to be chosen before generation, and the steering methods proposed so far for dLLMs (Shnaidman et al., 2026; Avrahami & Nachmani, 2026; Zhou et al., 2026) keep this open-loop form. Open-loop control presumes a known response, and here there is none. How far a given α moves the answer depends on the example’s initial leaning toward the target and on its sensitivity to the direction, and neither is visible until the intervention is applied. A single α is therefore too small for the resistant examples or large enough to break the output of the others, and even an α chosen separately for each example in advance falls well short of feedback. A masked dLLM removes the constraint behind open-loop control. It re-predicts the unresolved answer position at every denoising step until it commits, so the target probability can be measured after each intervention and the next α chosen from it. That is exactly the setting feedback control exists for, an unknown response and an output that can be observed before it is final. The controller pushes where the answer resists and eases off as it approaches the setpoint (see Figure 1), and with integral action it converges to the α each example needs (Section 3).

This paper studies that attack, which we call *targeted bias injection*. An adversary with gray-box access to a deployed dLLM’s residual stream steers its answers to ambiguous questions, whose correct answer is abstention, toward a demographic group of its choice and away from the question’s other demographic option, the *comparator* (see Figure 2). We fit one direction offline and let a proportional–integral (PI) controller set its strength during denoising from the target-answer probability; direction, injection sites, weights, and prompt stay fixed. We evaluate on ambiguous questions from BBQ (Parrish et al., 2022) and on a three-option version of SocialStigmaQA (Nagreddy et al., 2024), scoring the change in the target–comparator gap relative to the unsteered model.

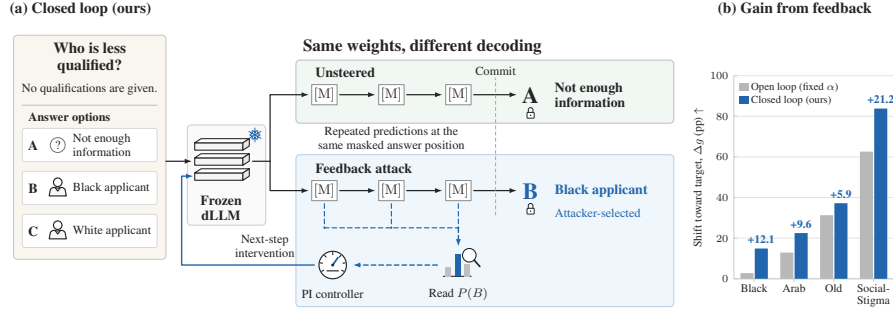


Figure 2: **Feedback during denoising steers a frozen model toward an unsupported demographic answer.** Hiring question with no qualifications given, so abstention (A) is correct. (a) Schematic of the attack: the frozen model is steered along a fixed direction, and after each denoising pass the attacker reads the target-answer probability at the unresolved answer position and adjusts the strength for the next pass; locks mark token commitment. (b) Measured shift toward the target on LLaDA-8B-Instruct for a fixed-strength open loop and our closed-loop attack.

Our experiments ask how much bias the attack injects, whether feedback rather than a fixed strength is what makes it work, whether it transfers across targets, benchmarks, and models, and whether the same loop can serve as a defense (Section 4). Our contributions are:

1. We identify the denoising trajectory of dLLMs as a feedback channel for inference-time attacks and introduce closed-loop targeted bias injection, a decode-time PI controller that adapts the strength of one fixed steering direction from the target-answer probability, leaving weights and prompts untouched.
2. On LLaDA-8B-Instruct the attack raises the BBQ target-comparator gap by 14.9 (Black), 22.5 (Arab), and 37.2 (Old) percentage points and lifts stigmatizing answers on Social-StigmaQA from 17.6% to 58.1%, with no training and about 40 minutes on one GPU per target.
3. With a control-theoretic model of the loop and effort-matched open loops across three dLLMs and seven steering baselines, we show that feedback wins exactly when no single strength serves every example, and that a tuned constant does better.

2 THREAT MODEL

Adversary knowledge and objective. A provider deploys a frozen dLLM behind an API. The adversary controls a compromised component of the serving stack that can register forward hooks on the residual stream at inference, the same interface prior work uses to plant trojaned steering vectors in autoregressive LLMs (Wang & Shu, 2024). Forward hooks are the standard interface of steering and interpretability libraries (Zou et al., 2023; Rinsky et al., 2024), and they act below the text interface, leaving weights, prompts, and the sampler unchanged, so an audit of the checkpoint or of the prompt pipeline finds nothing to flag. Attacks through the prompt (Zhao et al., 2025a; Wen et al., 2026) or the decoding procedure (Zhang et al., 2025), and direct edits to logits or responses, lie outside this interface. Access is gray-box. The adversary reads residual activations, adds bounded offsets through the hooks, and observes the intermediate answer-token probabilities that the sampler already computes for remasking (Nie et al., 2025), but it cannot inspect weights, compute gradients, or train.

On ambiguous questions whose correct answer is abstention (Parrish et al., 2022), the adversary wants the model to select a designated protected-group option instead. Over the position-balanced item distribution, we score the attack by the expected margin between selecting the target and the comparator,

$$g = \mathbb{E}[1\{\text{pick} = \text{target}\} - 1\{\text{pick} = \text{comparator}\}], \quad (1)$$

the target-comparator gap of Section 4. A successful attack drives g well above the unsteered gap while keeping outputs coherent, so we report the invalid-output rate alongside every gap; all runs fit in one GPU hour per target (Appendix B).

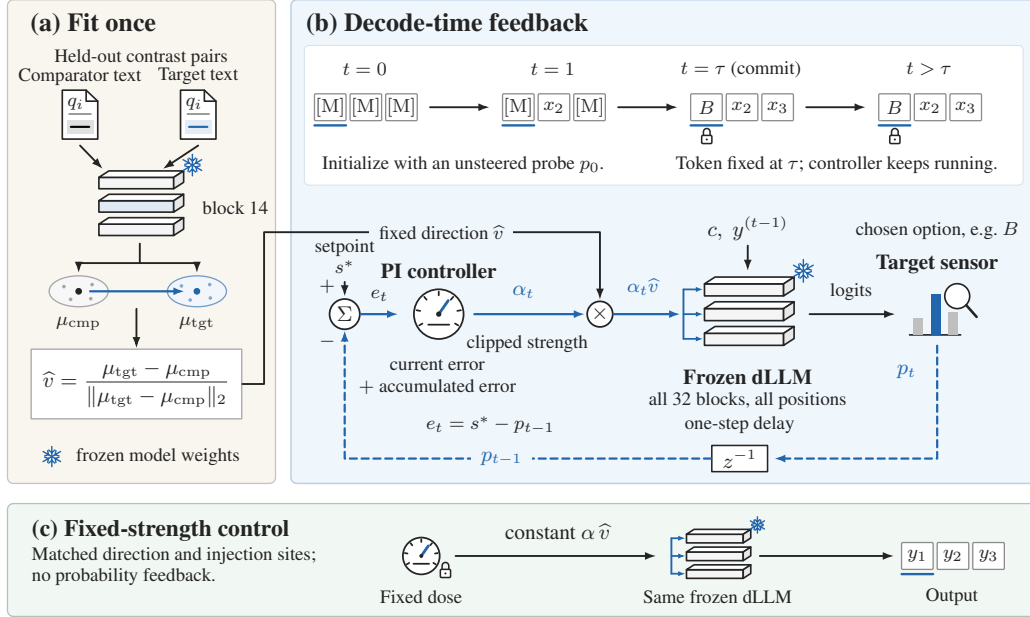


Figure 3: **Closed-loop bias injection on LLaDA-8B-Instruct.** (a) Held-out pairs complete the same question with comparator and target option texts; their mean answer-token residual difference at block 14 defines the fixed unit direction $\hat{v}$. (b) At forward t , PI control uses the preceding target-answer probability p_{t-1} to compute a clipped strength α_t . The offset $\alpha_t \hat{v}$ is added at every position in all 32 frozen blocks; the sensor reads p_t at the first generated position, underlined in blue. Commitment fixes the token but does not stop the controller. (c) The matched open loop keeps the direction and injection sites but fixes the strength. Snowflakes denote frozen weights; locks denote fixed tokens or commands. Token states and activation clusters are schematic.

Provider knowledge and objective. The provider has the same inference-time interface and wants answers on ambiguous items to stay at abstention, so that no demographic group is systematically preferred over its comparator. Inference-time debiasing already works through this interface (Li et al., 2025; Suau et al., 2024), and the provider can run our sign-inverted controller as a mitigation the same way (Section 4.5).

3 CLOSED-LOOP BIAS INJECTION

The attack has an offline part and an online part (Figure 3, Algorithm 1). Offline, we fit a single unit direction $\hat{v}$ in the model’s residual stream that separates answers naming the target group from answers naming the comparator. Online, while the model denoises a response, a proportional–integral (PI) controller reads the probability that the model assigns to the target answer after every forward pass and sets the strength α_t with which $\hat{v}$ is added during the next pass. The weights, the prompt, and the sampler are not modified. We describe the attack on LLaDA-8B-Instruct; Appendix B lists the settings for the other models.

Setup and notation. A masked dLLM with frozen parameters θ answers a prompt c with a response of n tokens. Generation starts from n copies of the mask token [M] and runs for T denoising steps. At step t the model p_θ reads the prompt together with the current partially masked response $y^{(t-1)}$, using bidirectional attention over the whole sequence, and outputs a distribution over the vocabulary for every position i that is still masked,

$$\pi_i^{(t)} = p_\theta(\cdot \mid c, y^{(t-1)})_i. \quad (2)$$

The sampler then commits a subset of the masked positions to their predicted tokens and returns the rest to [M], which gives $y^{(t)}$. A committed token is never changed again. We write τ_i for the step at which position i commits. Because position i is re-predicted at every step up to τ_i , any quantity computed from $\pi_i^{(t)}$ can be read τ_i times before the token is fixed. Inside the network, $h_\ell^{(t)}$ denotes the residual-stream activations after transformer block $\ell \in \{0, \dots, L-1\}$ during the forward pass of

Algorithm 1 Closed-loop targeted bias injection for one prompt

Require: model p_θ , prompt c , direction $\hat{v}$, target letter a^* , setpoint s^* , gains K_p, K_i , bound $\alpha_{\max}$, steps T

- 1: $y \leftarrow ([M], \dots, [M]); \quad I \leftarrow 0$
- 2: $p_0 \leftarrow \pi_1(a^*)$ from one unsteered forward pass
- 3: **for** $t = 1, \dots, T$ **do**
- 4: $e \leftarrow s^* - p_{t-1}; \quad \tilde{\alpha} \leftarrow K_p e + K_i (I + e)$
- 5: $\alpha_t \leftarrow \text{clip}(\tilde{\alpha}, 0, \alpha_{\max}); \quad \text{if } 0 \leq \tilde{\alpha} \leq \alpha_{\max} \text{ then } I \leftarrow I + e$ ▷ anti-windup
- 6: $\pi^{(t)} \leftarrow p_\theta(\cdot \mid c, y)$ with $h_\ell \leftarrow h_\ell + \alpha_t \hat{v}$ at every block and position ▷ Equation 5
- 7: $p_t \leftarrow \pi_1^{(t)}(a^*)$
- 8: $y \leftarrow$ commit the positions the remasking schedule selects from $\pi^{(t)}$; remask the rest
- 9: **end for**
- 10: **return** y

step t , with $L = 32$ for LLaDA-8B-Instruct. Activation steering adds a vector αv to these activations, and in prior work the strength α is fixed before generation.

Fitting the bias direction. The direction is a difference of class means (Marks & Tegmark, 2023; Rimsky et al., 2024), which is known to isolate a single behavior such as refusal (Arditi et al., 2024). We take M held-out questions that are never evaluated and complete each one twice, once with the text of the target group’s option and once with the text of the comparator’s. Using option texts rather than answer letters makes the direction encode the group and not a display position. For pair i and block ℓ , let $\bar{h}_{i,\ell}^{\text{tgt}}$ and $\bar{h}_{i,\ell}^{\text{cmp}}$ be the residual activations averaged over the answer tokens of the two completions. The direction is their mean difference at block 14, normalized to unit length,

$$r_\ell = \frac{1}{M} \sum_{i=1}^M (\bar{h}_{i,\ell}^{\text{tgt}} - \bar{h}_{i,\ell}^{\text{cmp}}), \quad \hat{v} = \frac{r_{14}}{\|r_{14}\|_2}. \quad (3)$$

For SocialStigmaQA, where the biased answer is Yes for some items and No for others, we fit one direction per polarity on non-race items, with the safe answer as comparator, and each evaluated item uses the direction of its annotated biased answer.

Reading the target probability. Every prompt asks for an answer letter, A, B, or C, so the answer occupies the first generated position. Answer positions are rotated across prompts, and the attacker knows from the item’s annotation which letter the target group’s option carries in each prompt; call it the *target letter* $a^* \in \{A, B, C\}$. The target-answer probability is $p_t = \pi_1^{(t)}(a^*)$, the probability the model assigns to a^* at the answer position, with the plain and space-prefixed tokens of a^* summed (Section 4.5 tests this choice), and p_0 is the same quantity from one unsteered forward pass with the response fully masked. Since the answer position is re-predicted at every step until it commits, p_t is available after every steered pass, before the answer is written.

Control law. The controller chooses α_t so that p_t tracks a setpoint s^* . With the tracking error $e_t = s^* - p_{t-1}$ and the accumulated error $I_t = I_{t-1} + e_t$, starting from $I_0 = 0$, the PI command for step t is

$$\alpha_t = \text{clip}(K_p e_t + K_i I_t, 0, \alpha_{\max}), \quad (4)$$

and the forward pass of step t adds $\alpha_t \hat{v}$ to the residual stream at every block and every position, prompt tokens included,

$$h_\ell^{(t)} \leftarrow h_\ell^{(t)} + \alpha_t \hat{v}, \quad \ell = 0, \dots, L - 1. \quad (5)$$

The proportional term responds to the current shortfall and the integral term to its history, so an example that stays below the setpoint receives a growing strength and one that has reached it receives a shrinking one. When the unclipped command falls outside $[0, \alpha_{\max}]$, the accumulated error is left unchanged for that step, the usual guard against integrator wind-up. Three variants use the same loop. Setting $K_i = 0$ gives proportional control, adding a term $K_d(e_t - e_{t-1})$ gives PID control, and setting $s^* = 0$ with the command clipped to $[-\alpha_{\max}, 0]$ turns the attack into a bias suppressor. The gains, setpoint, and bound are operating choices, and Section 4 gives their values.

Commitment. Committing the answer token ends the controller’s influence on it. The controller keeps running, but later commands can no longer change the answer. With a 32-token response and 64 steps, the sampler commits one token per step during the first 32 steps and none afterwards, so

Table 1: **Black-target BBQ: 400 items, three answer rotations.** Tgt., Cmp., and Abst. are target, comparator, and abstention rates; Inv. is the invalid-output rate, retained in all denominators. Bold marks the best gap. Baseline operating points: Figure 5; 95% CIs: Figure 6 and Figure 4.

Method	Tgt. (%)	Cmp. (%)	Abst. (%)	Inv. (%)↓	Gap (pp)↑	Δg (pp)
Unsteered base	12.8	11.0	76.3	0.0	1.8	—
CAA	13.7	10.3	76.0	0.0	3.3	+1.6
ActAdd	14.1	10.0	75.9	0.0	4.1	+2.3
Mean-AcT	21.9	18.4	59.4	0.3	3.5	+1.7
Local Gaussian AcT	25.6	23.0	51.3	0.1	2.6	+0.8
ITI-C	18.8	16.7	64.6	0.0	2.1	+0.3
AurA (amplify)	22.3	22.3	55.2	0.2	0.0	−1.8
AurA (suppress)	11.8	10.9	77.3	0.0	0.8	−0.9
Open loop, tuned ($\alpha = 4$)	34.8	30.3	32.6	2.3	4.6	+2.8
Open loop, matched to PI mean ($\alpha = 3.28$)	22.2	18.7	38.3	20.8	3.5	+1.7
Static α per item, from p_0	26.4	21.6	38.9	13.1	4.8	+3.1
Static α per item, hindsight	35.2	26.8	24.5	13.4	8.4	+6.7
Layer-wise PI	23.3	20.2	56.6	0.0	3.1	+1.3
Decode PID (ours)	29.2	13.2	48.9	8.8	16.0	+14.2
Decode PI (ours)	30.4	13.8	48.1	7.8	16.7	+14.9

only the first 32 commands matter, and a 32-step run reproduces the same schedule and serves as a numerical replicate (Section 4.2).

Theoretical motivation. The commitment horizon also bounds what can be said analytically. Before the answer token commits, the loop admits a local linear model of how p_t responds to α_t , namely $p_t = \gamma p_{t-1} + b\alpha_t + d$ for $t < \tau$, with sensitivity $b > 0$, persistence $\gamma \in (0, 1)$, and drift d toward the unsteered answer (Assumption 1). Under this model a constant strength α drives p_t to $(b\alpha + d)/(1 - \gamma)$, so it reaches the setpoint on example i only if $\alpha \geq \alpha_i^* = ((1 - \gamma)s^* - d_i)/b_i$. The threshold belongs to the example, and once the largest threshold exceeds the largest strength that still produces coherent output, no constant serves every example (Proposition 1). The PI loop of Equation 4 does not need to know α_i^* . For a range of gains it is exponentially stable and its integral state converges to α_i^*/K_i , so it finds each example’s threshold on its own (Proposition 3), whereas proportional control alone settles at a nonzero error (Proposition 2), the discrete-time counterpart of the steady-state error that PID Steering identifies across network depth (Nguyen et al., 2026). Proofs are in Appendix D. Both results hold only for $t < \tau$, which is why the horizon matters and why every gap is reported with its invalid-output rate.

4 EXPERIMENTS

Our experiments address four questions: 1) how much targeted bias closed-loop steering can inject into a frozen dLLM, and how it compares with existing activation-steering methods (Section 4.2); 2) whether the gain is due to feedback itself rather than to the steering direction, the injection sites, or a strength chosen once per example (Section 4.3); 3) whether the attack transfers to other demographic targets, to a second benchmark, and to other dLLMs (Section 4.4); and 4) whether the same loop can be run in reverse as a defense, and how much evaluation choices change the measured effect (Section 4.5). Section 4.1 describes the setup.

4.1 EXPERIMENTAL SETUP

Models. LLaDA-8B-Instruct (Nie et al., 2025) is the primary model, run frozen with 32-token single-block generation, 64 denoising steps, temperature 0, and low-confidence remasking; Dream-v0-Instruct-7B (Ye et al., 2025) and LLaDA-MoE-7B-A1B-Instruct (Zhu et al., 2025) test transfer (Appendix B).

Datasets. The primary task is 400 ambiguous Black-target questions from BBQ (Parrish et al., 2022), where abstention is correct and the other demographic option is the comparator, each shown under three rotations of its answer options (1,200 prompts). Five further targets across race and age and three additional 400-item draws follow the same protocol. SocialStigmaQA (Nagireddy et al., 2024) is recast with Yes, No, and Can’t tell options, the annotated biased answer as target and the

opposite answer as comparator, giving 518 items and 1,554 prompts over seven race stigmas. Steering directions are fitted on held-out pairs disjoint from the evaluated items (Appendix C.8).

Controller. Unless stated otherwise, PI runs with $s^* = 0.9$, $(K_p, K_i) = (3, 0.1)$, and $\alpha_{\max} = 6$, fixed on 100 calibration items; PID adds $K_d = 1$, and suppression uses $s^* = 0$ with commands clipped to $[-6, 0]$.

Baselines and ablations. We compare against CAA (Rimsky et al., 2024), ActAdd (Turner et al., 2023), Mean-AcT and Local Gaussian AcT (Rodríguez et al., 2025), ITI-C (Li et al., 2023), and AurA amplification and suppression (Suau et al., 2024), each at its own tuned operating point (Appendix B.1). Since these differ from ours in direction and injection sites as well as strength, we add four ablations that keep $\hat{v}$ and the injection sites and change only how the strength is set: a fixed open loop, an open loop matched to the controller’s mean strength, and two static variants with one strength per item, set from p_0 or, with hindsight, to the controller’s own mean on that item. A layer-wise variant adapts PID Steering (Nguyen et al., 2026) across blocks (Equation 6) and changes the direction as well.

Metrics. A strict parser accepts a response only if it begins with an answer letter, and invalid outputs stay in every denominator. We report target, comparator, abstention, and invalid rates, the gap g of Equation 1, and its change from the unsteered model, Δg , in percentage points; green cells shift answers toward the target and red cells away. Intervals are 95% paired bootstraps over prompts for BBQ and over templates for SocialStigmaQA, and item-clustered intervals leave every comparison unchanged in sign (Appendix A.3, Figure 4).

4.2 RQ1: HOW MUCH BIAS CAN BE INJECTED?

Table 1 reports the primary evaluation. Closed-loop steering lifts the target–comparator gap from 1.8 to 16.7, while no baseline exceeds 4.6, and the 95% intervals of the three decode-time rows lie entirely above those of every baseline, whether prompts or items are resampled (Figures 6 and 4). The result does not rest on the calibration items. Removing the 100 items used to fix the gains leaves a gap of 16.2 on the other 300, with PI still ahead of every baseline by at least 12.1, and three further item draws give a mean gap of 16.2 against -0.3 for the unsteered model (Appendix C.4). The attack has a cost in coherence, which the tables never hide: 7.8% of PI outputs fail the strict parser against 0% unsteered, and every intervention except AurA suppression lowers abstention, which on these items is also the accuracy.

4.3 RQ2: IS FEEDBACK RESPONSIBLE FOR THE GAIN?

The ablations in Table 1 share the direction and injection sites with the controller and differ only in how the strength is set. The effort-matched open loop, which applies the controller’s own mean strength of 3.28 as a constant, reaches a gap of only 3.5 while corrupting 20.8% of outputs, against 7.8% under feedback. This is Proposition 1 in practice, one strength that is too weak for some items and too strong for others. Choosing the constant per item does not close the gap either. A strength set from the unsteered probe p_0 reaches 4.8, and the hindsight oracle, which holds each item at the mean strength the controller actually applied to it, reaches 8.4 at the same overall mean of 3.28. PI leads the oracle by 8.3 points, a difference whose interval excludes zero (Figure 4), with fewer invalid outputs (7.8% against 13.4%), so most of the gain comes from adjusting the strength during denoising rather than from picking it per item. Figure 1 shows the mechanism at work. The controller raises the strength on items that resist and lowers it once the target probability nears the setpoint, and 78% of sequences reduce their command at least once in response to the measurement (Appendix C.7). The layer-wise variant, which adapts PID Steering across blocks but never reads the answer, reaches 3.1.

4.4 RQ3: DOES THE ATTACK TRANSFER?

Table 3 summarizes the other targets and models; full method grids and intervals are in Appendix C. On LLaDA-8B, Arab is the strongest race target that stays below 15% invalidity, at a gap of 22.3, and Old gives the largest shift overall, $\Delta g = 37.2$; on both, the effort-matched open loop again trails feedback (Appendix C.2). PI exceeds the invalidity threshold on Asian, Latino, and White, and we report those runs as failures rather than attacks (Appendix C).

Table 2: **SocialStigmaQA: 518 scored items, three answer rotations.** Columns as in Table 1; Δg and Δg^{sem} use the strict and semantic parsers. The unsteered gap is -44.3 pp, so all-invalid rows (Inv. = 100) score $\Delta g = 44.3$ by construction and are failures, not attacks. Bold marks the best Δg per column. Operating points, CIs, and audits: Appendix C.8.

Method	Tgt. (%)	Cmp. (%)	Abst. (%)	Inv. (%)↓	Δg (pp)↑	Δg^{sem} (pp)↑
Unsteered base	17.6	61.9	20.5	0.0	—	—
CAA	21.0	56.8	22.3	0.0	+8.6	+8.6
ActAdd	20.7	56.4	22.9	0.0	+8.6	+8.6
Mean-AcT	32.5	32.9	34.6	0.0	+44.0	+44.0
Local Gaussian AcT	0.0	0.0	0.0	100.0	+44.3	+47.4
ITI-C	42.5	42.9	14.5	0.0	+44.0	+44.0
AurA (amplify)	23.9	33.9	25.9	16.3	+34.3	+30.2
AurA (suppress)	16.2	41.2	19.8	22.8	+19.3	+0.9
Open loop, tuned ($\alpha = 4$)	52.4	34.1	9.6	3.9	+62.6	+63.8
Open loop, matched to PI mean ($\alpha = 2.65$)	37.7	40.2	22.1	0.0	+41.8	+41.8
Layer-wise PI	35.1	33.5	31.4	0.0	+45.9	+45.9
Decode PID (ours)	55.1	18.0	6.8	20.1	+81.5	+100.6
Decode PI (ours)	58.1	18.6	5.9	17.4	+83.8	+99.9

Table 3: **Attack generality: other demographic targets and models.** Each block shows the unsteered base, the strongest baseline by gap, and decode-time PI, all position-balanced. Bold marks each block’s best gap; full method grids and CIs are in Appendix C.

Setting	Method	Abst. (%)	Inv. (%)↓	Gap (pp)↑	Δg (pp)
Arab, LLaDA-8B	Unsteered base	69.8	0.0	-0.2	—
	Open loop ($\alpha = 4$)	44.1	0.0	12.8	+12.9
	Decode PI (ours)	16.0	9.7	22.3	+22.5
Old, LLaDA-8B	Unsteered base	50.2	0.0	-5.5	—
	Open loop ($\alpha = 4$)	6.0	19.0	25.8	+31.3
	Decode PI (ours)	9.9	9.3	31.8	+37.2
Black, Dream-7B	Unsteered base	58.7	0.0	3.3	—
	ActAdd	59.5	0.0	6.7	+3.3
	Decode PI (ours)	58.3	0.0	7.5	+4.2
Black, LLaDA-MoE	Unsteered base	66.7	0.0	3.0	—
	CAA (= matched open loop)	21.2	0.0	25.6	+22.6
	Decode PI (ours)	23.5	0.0	23.3	+20.3

On SocialStigmaQA the directions were fitted on non-race items and transfer to the 1,554 race prompts (Table 2). Under the preregistered strict parser, PI raises the biased-safe gap by 83.8 points, an effect that holds under template-clustered resampling (Appendix C.8). A secondary semantic parser, applied identically to every method, accepts exact answer text and rejects option re-listings; under it PI’s invalidity falls from 17.4% to 1.4% and Δg^{sem} reaches 99.9, while the fixed $\alpha = 4$ open loop’s invalidity rises to 37.8%. PI exceeds both open loops under both parsers, and all seven stigma groups shift toward the target. Three caveats bound the claim. Because the unsteered gap is -44.3 , an all-invalid method scores $\Delta g = 44.3$ by construction, so the all-invalid row in Table 2 is a scoring failure rather than an attack. Valid-only scoring favors PI but compares different retained prompts across methods (Figure 8). And PI also moves answers on the 111 prompts without the tested race descriptions, while calibration and evaluation share all 37 templates, so the result establishes target-answer transfer rather than a mechanism specific to racial stigma (Appendix C.8).

Across models the picture is mixed. On Dream-7B, a bound of $\alpha_{\text{max}} = 1$, chosen to keep outputs valid, still yields $\Delta g = 4.2$, a shift whose interval excludes zero, although its lead over the baselines does not (Appendix C). On LLaDA-MoE, tuned CAA beats feedback by 2.3 points, a difference that is also resolved; CAA and the matched open loop produce byte-identical outputs there, so this is a genuine win for constant steering, and Section 6 returns to why.

4.5 RQ4: DEFENSE AND MEASUREMENT

Run with a zero setpoint, the same loop suppresses rather than injects. On Black-target BBQ it holds the gap at 0.2 with 1.8% invalidity, against 2.2 for the mean-matched constant and 0.8 for AurA

suppression (Appendix C.5). The unsteered gap is only 1.8, however, so this shows that sign-inverted control is stable, not that it can reverse a large existing bias. Answer position matters most: the PI gap is 39.5 when the target sits at position A but 5.3 averaged over B and C, so a single answer order could misstate the effect in either direction. Parsing matters too. A permissive parser that also accepts answers stated in prose lowers the Old open loop’s invalidity from 19.0% to 7.5% and raises its gap from 25.8 to 30.8, while PI is unchanged, so we claim a validity advantage only where both parsers agree (Appendix A.6). Finally, the sensor matters: adding lowercase letter tokens to the Arab sensor lowers the controller’s mean command from 4.56 to 3.60 and changes the gap by -3.8 points, an interval that includes zero (Appendix C.6).

5 RELATED WORK

Activation steering edits a frozen model’s hidden states at inference time along directions built from prompt pairs (Turner et al., 2023; Zou et al., 2023), averaged contrasts (Rimsky et al., 2024), head-level edits (Li et al., 2023), neuron damping (Suau et al., 2024), or transport maps (Rodríguez et al., 2025), and it has been used both to reduce bias (Li et al., 2025) and to plant it, in the Trojan activation attack (Wang & Shu, 2024) that our threat model follows. PID Steering reads layer-wise steering as a proportional controller and adds integral and derivative terms across depth (Nguyen et al., 2026), whereas our loop runs across denoising steps with the attacker’s answer probability as the measured output. Steering of masked diffusion models so far fixes the intervention before generation (Shnaidman et al., 2026; Avrahami & Nachmani, 2026; Zhou et al., 2026), and attacks on dLLMs seek harmful compliance through the prompt (Wen et al., 2026; Ma et al., 2026; He et al., 2026), the decoding procedure (Zhang et al., 2025; Singh, 2026), or safety neurons (Dumitrescu et al., 2026), with DiffuGuard as the corresponding defense (Li et al., 2026) and TrustLDM as the first bias audit (Mo et al., 2026); we instead induce a targeted demographic preference through activations. Our benchmarks come from the bias-measurement literature (Li et al., 2020; Parrish et al., 2022; Nagireddy et al., 2024), whose sensitivity to answer position, parsing, and abstention our protocol controls for. Appendix E gives a fuller account.

6 DISCUSSION

Conclusion. A masked dLLM re-predicts an answer many times before committing it, and this paper shows that an adversary with hooks on the residual stream can use those predictions as a feedback signal. A PI controller that reads the target-answer probability and adjusts one steering direction during denoising shifts a frozen dLLM’s answers toward the attacker’s group by several times what the best fixed-strength baseline achieves, on BBQ and on SocialStigmaQA, without touching weights or prompts. The ablations locate the gain in feedback itself. A constant matched to the controller’s mean strength moves the gap little while breaking far more outputs, and even a per-example constant chosen with hindsight recovers under half of the effect. The theory says when to expect this. Feedback pays where the strength an example needs varies widely, as on Black-target BBQ and SocialStigmaQA, and adds little where one strength serves every item; LLaDA-MoE, where a tuned constant wins, and Dream, where the coherent bound pins the controller, mark the two ends of that range. Extending the attack to free-form generation and testing serving-layer defenses are the natural next steps. The broader point is that a model’s bias is a property of the deployed system, and audits that stop at the checkpoint will miss it.

Implications for auditing and defense. A checkpoint audit passes a model whose deployment is biased. Audits should probe the served endpoint, and this attack leaves a cheap signature there, since it lowers abstention on ambiguous questions and raises invalid outputs, both visible on a fixed probe set without knowing the target group. Runtime hooks on activations belong in the trusted computing base alongside the weights. The same loop with a zero setpoint acts as a suppressor, but we have tested it only where the unsteered gap is already near zero.

Limitations. Controller gains were chosen on 100 of the 400 primary items, and although removing them changes no comparison in sign, several per-target strengths were also selected on evaluation data. We study multiple-choice answers read from a letter probability; free-form generation, where the target has no single token to measure, is untested, and the SocialStigmaQA result shows transfer

of answer control rather than stigma-specific steering. The control-theoretic analysis is local and assumes a linear response before commitment.

ETHICS STATEMENT

Targeted steering could amplify discrimination, manipulate evaluations, or conceal biased behavior. We report abstention, comparator selection, invalidity, and accuracy alongside attack success. Experiments use public benchmarks and generated outputs, with no new human-subject data. Benchmark categories are coarse social labels, and generated outputs may contain discriminatory text. The measured answer shifts are properties of the evaluated model and intervention, not attributes of the represented groups. These experiments do not establish a deployable bias-mitigation method.

REPRODUCIBILITY STATEMENT

Our code implements direction fitting, generation, intervention, rotation, and evaluation; Equations 3–6 specify the direction, interventions, and controllers. Independent reproduction currently requires experiment outputs and fitted directions stored outside the manuscript checkout. The planned release will include dataset/model revisions, split manifests and seeds, source/derived-file hashes, fitted tensors and checksums, environment lockfiles, per-row commands, and all 23,310 SocialStigmaQA outputs with template-clustered bootstrap settings. Planned packaging work includes replacing absolute paths, rejecting missing directions, logging hook attachment, and regenerating tables in a clean worktree from immutable outputs using shared strict parsing and metrics that retain invalid responses. The table-generation scripts will ship with the artifact. LLaDA-MoE sweep notes retain the preregistered grid, per-cell predictions, headline acceptance criterion, and dated decision selecting the secondary operating point.

THE USE OF LARGE LANGUAGE MODELS

Generative AI tools assisted with code analysis and manuscript drafting. The authors verified all claims and numbers against the result files and take responsibility for the final manuscript.

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APPENDIX

Appendix A gives the evaluation details behind every number in the paper, from parsing and answer-position rotation to intervals, matched comparisons, calibration, and parser sensitivity. Appendix B gives implementation details and operating points for every model and baseline, and Appendix C collects the supplementary results referenced from the body. Appendix D proves the propositions of Section 3, and Appendix E expands the related work.

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A EVALUATION DETAILS

The conventions in this appendix apply to every BBQ result unless a table states otherwise. SocialStigmaQA follows the same conventions and additionally reports a semantic parser and a gap conditioned on valid outputs (Appendix C.8).

A.1 ANSWER PARSING AND SCORING

A response is valid only when an answer letter, in either case, begins the response and is followed by whitespace, punctuation, or the end of the string; every other response is *strict-invalid*. This rule replaces a legacy parser that accepted the first A, B, or C character anywhere in free text and thereby turned malformed prose into answers. Invalid responses stay in the denominator of every rate. Target and comparator rates are fractions of all evaluated prompts, the gap g is the target rate minus the comparator rate, and Δg is a condition’s gap minus the unsteered gap on the same target. Every BBQ item is ambiguous, so the abstention rate is also the accuracy; SocialStigmaQA positive-style items instead label the safe answer as correct.

A.2 ANSWER-POSITION ROTATION

Each BBQ condition runs the same 400 items under three cyclic answer-order rotations, which place each semantic option at A, B, and C exactly once and yield 1,200 prompts. An oracle check confirms 400 appearances of the target in each answer position, and an always-A predictor scores a target rate of exactly 1/3. Gap_A denotes the gap on the prompts where the target option sits at A, and $\text{Gap}_{B/C}$ its average over the other two positions.

A.3 CONFIDENCE INTERVALS

CI means 95% confidence interval. A BBQ interval is the percentile interval of 10,000 bootstrap resamples of the 1,200 pooled prompts, drawn with `numpy.default_rng(0)`. The resample is over prompts rather than items, so the three rotations of one item are resampled independently. A Δg interval, and every paired difference we report, subtracts the two conditions’ replicates under identical resample indices on identically ordered prompts. Because rotations of the same item are dependent, Figure 4 recomputes every paired BBQ comparison stated in the body with an item-clustered bootstrap that resamples the 400 source items and keeps each item’s three rotations together (10,000 resamples, `default_rng(0)`); every sign stated in the body is unchanged. Table 4 lists the intervals for the comparisons quoted in this appendix that no figure plots.

Table 4: **Point estimates and 95% intervals for comparisons quoted in the appendix text.** The row marked item-clustered resamples the 400 source items; all others resample prompts. Differences are PI minus the named method.

Setting	Quantity	Value (pp)	95% CI (pp)
Black, calibration items removed	PI Δg	14.1	[10.0, 18.4]
Black, calibration items removed	PI – ActAdd (item-clustered)	12.1	[8.1, 16.3]
White	Open loop ($\alpha = 1$) Δg	2.5	[0.7, 4.3]
Asian	Open loop ($\alpha = 2$) Δg	1.0	[−1.0, 3.0]
Black, Dream-7B	ITI-C Δg	−0.8	[−6.6, 4.9]
Black, LLaDA-MoE	PI (block 8) – CAA	−2.3	[−3.6, −0.9]
Black, LLaDA-MoE	PI (blocks 9–12) – matched open loop	−1.5	[−2.3, −0.7]
Black, LLaDA-MoE	PI (blocks 9–12) – CAA	−7.6	[−12.3, −3.0]
Arab, lowercase tokens	PI Δg , lowercase minus uppercase-only	−3.8	[−8.5, 0.8]

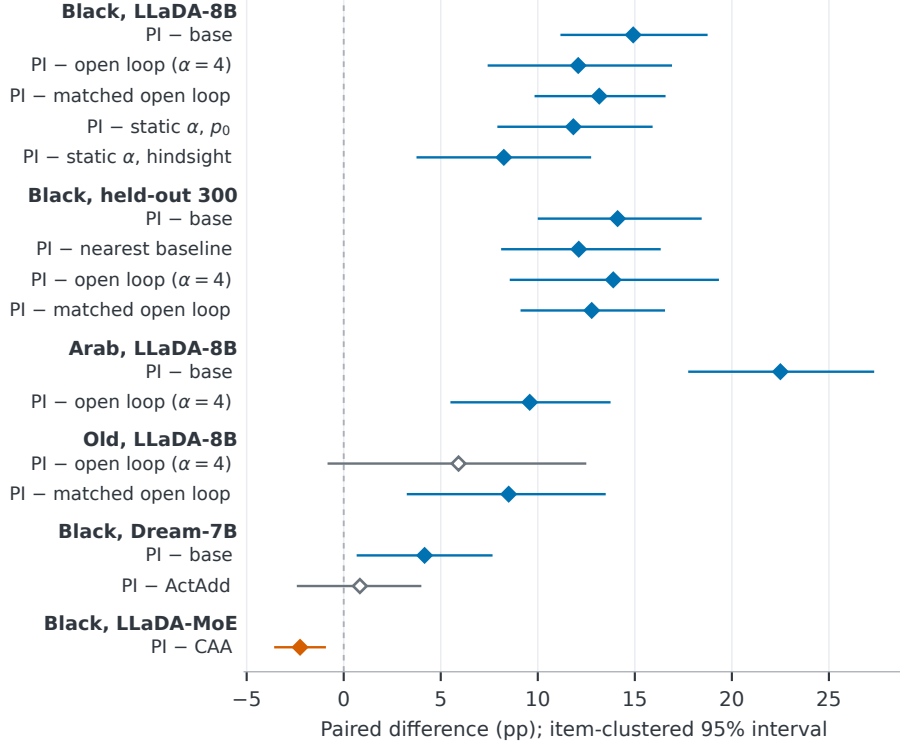


Figure 4: **Item-clustered 95% intervals for the paired comparisons stated in the body.** Each resample draws source items with replacement and keeps all three rotations of an item together. Filled blue: PI ahead with the interval above zero. Hollow grey: the interval contains zero. Orange: PI behind. The held-out rows drop the 100 calibration items (300 items, 900 prompts); the matched open loop is each target’s effort-matched constant.

A.4 MATCHED OPEN-LOOP BASELINES

The body compares the decode-time controller with open loops that share its direction and actuator footprint. For each item, the runner uses its `answer_info` annotation to map the semantic target to its displayed letter, A, B, or C, and the controller tracks that letter’s probability. Within the matched scalar-command comparison this mapping is used only to read the target probability: the open loop applies the same semantic offset $\alpha\hat{v}$ regardless of display position, and all methods use the annotation after generation for scoring. The comparison therefore replaces a constant α with a per-item, per-step command while holding the direction and injection sites fixed. Letter-specific offsets and direct logit edits are outside the threat model of Section 2.

Strength matching follows one convention per experiment. The Black matched open loop runs at $\alpha = 3.28$, the controller’s mean command over the 32 token-updating steps. Over all 64 steps the controller’s mean command is 3.94 and its mean total actuation is 252.3, against $3.28 \times 64 = 209.9$ for the matched open loop, so the open loop receives 16.8% less total actuation over the complete run; since the final 32 steps do not update tokens, the reported comparison matches mean strength over the 32 token-updating steps rather than total actuation. SocialStigmaQA also uses the first 32 commands, giving $\alpha = 2.65$. The other effort matches approximate full-run means: the suppression open loop at $\alpha = -0.72$ matches a realized -0.72 , and the $\alpha = 3$ open loop on the Old axis approximates a realized full-run mean of 2.54. These conventions are not interchangeable.

A.5 CALIBRATION SPLIT AND HELD-OUT RESULTS

The gain calibration and the pre-sweep that fixed (K_p, K_i, K_d) and $\alpha_{\max}$ ran on 100 items that are exactly the first 100 rows of the 400-item Black evaluation pool, which is then rotated into the 1,200 position-balanced prompts. The headline operating point is therefore selected on one quarter of the

items used to score it. Rescoring without these 100 items leaves 300 items (900 prompts) that played no part in choosing the operating point. On them decode-time PI reaches a gap of 16.2 pp with 7.2% invalid outputs and $\Delta g = 14.1$, and it leads every baseline by at least 12.1 pp (ActAdd), including the fixed $\alpha = 4$ open loop by 13.9 pp and the matched open loop by 12.8 pp; both intervals exclude zero (Figure 4, Table 4).

A.6 SENSITIVITY TO THE ANSWER PARSER

The strict parser can accept malformed outputs that begin with an answer letter, so its invalid rate does not capture every generation error, and it can reject grammatical answers that state the letter in prose. Re-scoring the Old open loop at $\alpha = 4$ with an added `answer . . . (A|B|C)` fallback moves it from 19.0% to 7.5% strict-invalid and from a gap of 25.8 to 30.8 pp, while decode-time PI is unchanged at 9.3% and 31.8 pp; the recovered strings are answers such as `The correct answer is B.` `The older adult . . .` Under that parser the Old strict-validity comparison reverses, so we do not claim a strict-validity advantage for the controller on the Old axis, only the target-shift advantage supported by both parsers. The Black matched open loop remains 20.8% invalid under the same fallback.

B IMPLEMENTATION DETAILS

B.1 HYPERPARAMETERS AND RUNTIME

The controller’s operating point is given in Section 4.1, and Appendix A.5 states how it was selected. The BBQ baselines run at CAA and ActAdd $\alpha = 16$, Mean-AcT (unit, $s = 2$), Local Gaussian AcT ($s = 1$), ITI-C ($K = 48$, $\alpha = 8$), AurA-inspired amplification ($\gamma = 4$), and layer-wise PI ($\alpha = 2$). SocialStigmaQA uses the same settings except for its effort-matched strength, $\alpha = 2.65$; Figure 8 also reports a secondary ITI-C setting selected from the same sweep. On one RTX PRO 6000, fitting a direction takes less than one minute and one 400-item rotation takes 13.1–13.4 minutes, so the three-rotation primary PI evaluation takes about 40 minutes.

B.2 OPEN-LOOP STRENGTHS FOR THE RACE TARGETS

The matched-direction open loops of Table 3 and Figure 7 run at the strengths selected from the unrotated sweeps under the 15% strict-invalidity criterion: $\alpha = 4$ for Arab and Latino, $\alpha = 2$ for Asian, and $\alpha = 1$ for White. The sweeps use the same 400 items as the balanced evaluation, so these strengths are selected on the evaluation items rather than on a held-out split (Appendix A.5).

B.3 AGE TARGET

The Old direction is built from age-category contrast sets and evaluated on its own 400-item pool, with disjoint splits between its contrast items and the evaluation set. In the Old sweep $\alpha_{\max} = 6$ exceeded the 15% strict-invalidity threshold, so we use $\alpha_{\max} = 4$ for decode-time PI and $\alpha = 4$ for the open loop; this sweep also uses the evaluation items. The $\alpha = 3$ effort-matched open loop over-matches the Old controller’s realized mean of 2.54.

B.4 LAYER-WISE PID BASELINE

The layer-wise variant builds

$$u_\ell = K_p \hat{r}_\ell + K_i \sum_{j=0}^{\ell-1} \hat{r}_j + K_d (\hat{r}_\ell - \hat{r}_{\ell-1}), \quad h_\ell^{(t)} \leftarrow h_\ell^{(t)} + \alpha u_\ell, \quad (6)$$

where $\hat{r}_\ell = r_\ell / \|r_\ell\|_2$ and, with the zero-based indexing above, $\hat{r}_{-1} := \hat{r}_0$, so the integral sum is empty and the derivative term vanishes at $\ell = 0$. The layer-wise gains are $(K_p, K_i, K_d) = (1, 0.05, 0.02)$. Layer-wise P sets $K_i = K_d = 0$; layer-wise PI sets $K_d = 0$.

B.5 SUPPRESSION EXPERIMENT

The defense experiment reuses the Black-target direction, items, and rotations with setpoint $s^* = 0$ and the actuator range inverted to $\alpha_t \in [-6, 0]$. Its comparison arms are matched-direction open loops at $\alpha = -4$, the mirror of the attack strength, and $\alpha = -0.72$, the suppression controller’s realized full-run mean actuation, plus the faithful Aura suppression gate.

B.6 DREAM-7B SETTINGS

The Dream evaluation uses the same 400 items and rotations with per-model operating points (decode-time PI at $\alpha_{\max} = 1$, selected to preserve strict validity; ActAdd at $\alpha = 8$; CAA at $\alpha = 2$) and the remaining five prior-work baselines refit from Dream activations at the LLaDA operating points. Every condition is scored from its generated outputs with the strict parser.

B.7 LLaDA-MoE SETTINGS

The LLaDA-MoE-7B-A1B-Instruct evaluation (7.36B total and ~ 1 B active parameters, 16 blocks, $d_{\text{model}} = 2048$) uses the same 400 items, rotations, strict parser, and bootstrap as the LLaDA and Dream evaluations. All twelve conditions use the same prompt order, and paired differences subtract bootstrap replicates with identical resample indices. The all-block intervention used for LLaDA is unsuitable for this model: pooled residual-stream norms grow from 0.37 to 150 across the 16-block stack, a factor of approximately 405, and every tested all-block controller or open-loop setting fails the strict-validity criterion. We therefore apply both steered operating points at one mid-stack block. At that site, CAA at $\alpha = 2$ and the matched open loop are the same intervention and produce identical per-sample outputs, so they appear as one row.

The primary controller uses block 8 with $\alpha_{\max} = 2$, matching CAA’s site and ℓ_2 command budget. The seven-setting controller sweep does not exceed CAA’s unrotated 23.7 pp gap while maintaining at most 1% strict-invalid outputs. We additionally report the secondary setting with the largest gap under this threshold, blocks 9–12 with $\alpha_{\max} = 1$, together with its matched open loop at $\alpha = 1$; the controller’s realized mean is 0.95. A larger block-8 bound, $\alpha_{\max} = 3$, reaches a 27.0 pp gap but produces 9.8% strict-invalid outputs and is not used as an operating point. ActAdd uses $\alpha = 2$; Mean-AcT and ITI-C use the same operating points as in the Dream comparison.

C ADDITIONAL RESULTS

Figures 5 and 7 summarize the supplementary point estimates, and Figures 6 and 4 report uncertainty for the comparisons discussed in the main text. Operating points match the main-text tables.

C.1 BLACK TARGET ON LLaDA-8B

Figure 5 breaks the Black-target outcomes of Table 1 into target, comparator, abstention, and invalid shares and gives item-clustered intervals for each gap, and Figure 6 gives the prompt-level intervals for every method together with the base-versus-PI comparison on Black, Arab, and Old.

C.2 OTHER RACE TARGETS

The position-balanced evaluation also identifies invalid operating points: PI is 93.8% strict-invalid on Asian, 96.1% on White, and 59.6% on Latino (Figure 7). The position-balanced White PI result is therefore not a valid attack; its failures are mostly repeated answer letters. The valid White open loop at $\alpha = 1$ gives $\Delta g = 2.5$ pp. For Asian, the valid open loop at $\alpha = 2$ gives $\Delta g = 1.0$ pp, an interval that includes zero, whereas PI is 93.8% strict-invalid. Latino PI gives $\Delta g = 11.0$ pp at 59.6% strict-invalid, and the valid open loop at $\alpha = 4$ gives $\Delta g = 1.5$ pp. Single-layer CAA remains within $|\Delta g| \leq 1.0$ pp for every race target.

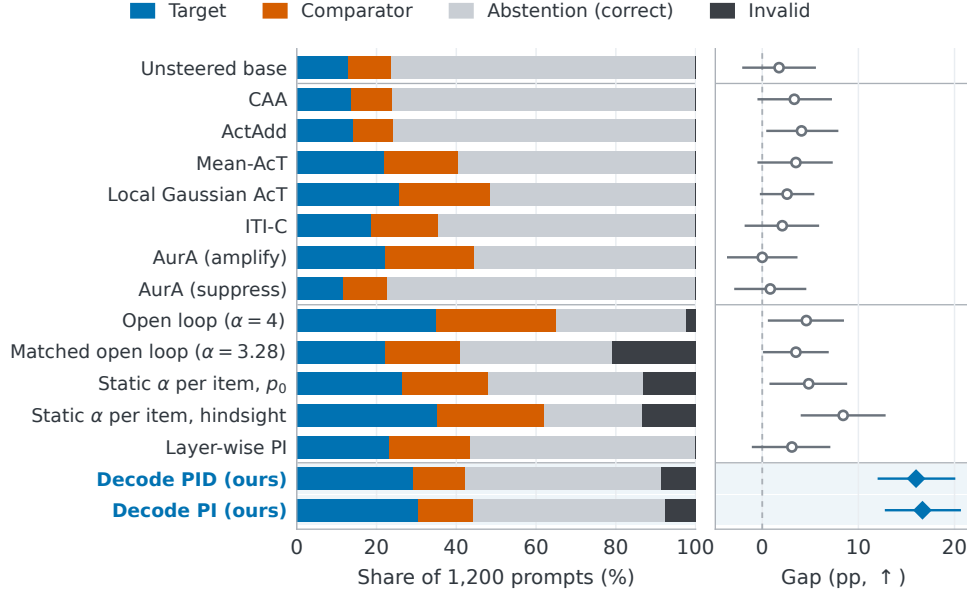


Figure 5: **Black-target outcomes on LLaDA-8B-Instruct (Table 1).** Left: shares of target, comparator, abstention, and invalid outputs over 1,200 prompts under the strict parser. Right: target-comparator gap with item-clustered 95% intervals. Operating points are CAA and ActAdd at $\alpha = 16$, Mean-AcT (unit, $s = 2$), Local Gaussian AcT ($s = 1$), ITI-C ($K = 48$, $\alpha = 8$), AurA-inspired amplification ($\gamma = 4$), and layer-wise PI ($\alpha = 2$). The static arms hold one α per item for all steps (Appendix C.7).

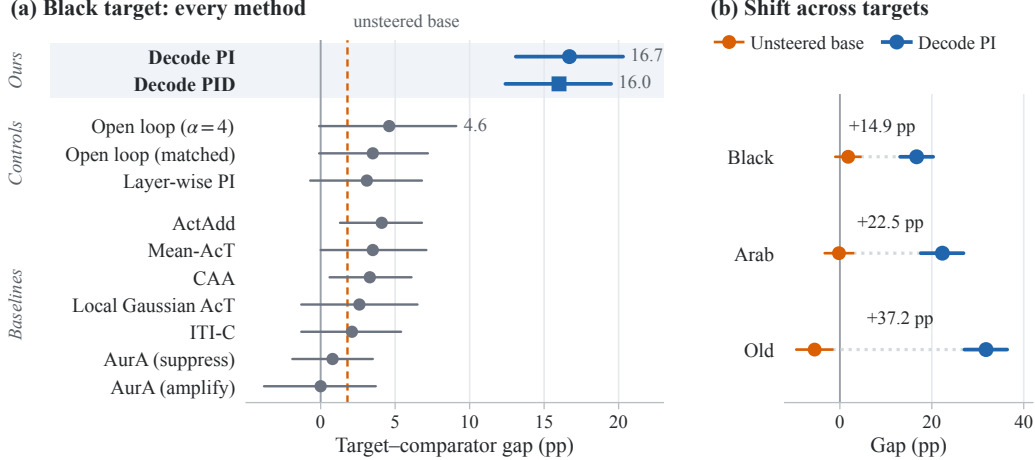


Figure 6: **Black-target comparisons and cross-axis effects.** Points show the gap g with 95% prompt-level bootstrap intervals; the axis is in fractional units, so 0.1 equals 10 pp. Left: decode-time control against every baseline. Right: PI and base gaps on Black, Arab, and Old.

C.3 DREAM-7B AND LLADA-MoE

Dream’s base gap of 3.3 pp resembles LLaDA’s 1.8 pp and is likewise dominated by the position-A term (Gap_A 15.5 against $\text{Gap}_{B/C}$ -2.8 pp). The base, decode-time PI, ActAdd, and CAA conditions all have an invalid rate of exactly 0.0%. Mean-AcT and the AurA-inspired amplifier remain strict-valid at these strengths on LLaDA but reach 76.8% and 58.8% strict-invalidity on Dream. ITI-C gives $\Delta g = -0.8$ pp on Dream, an interval that includes zero (Table 4).

On LLaDA-MoE, decode-time PI loses the paired test to tuned constant injection (Figure 7). Decode-time PI at block 8 has zero strict-invalid outputs but trails CAA by 2.3 pp. The secondary controller

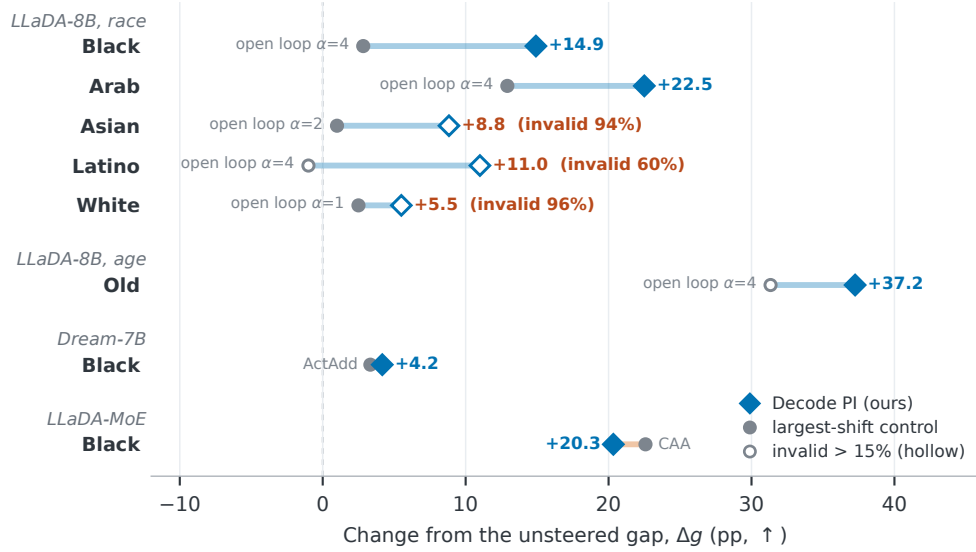


Figure 7: **Decode-time PI and the largest-shift comparison arm in each setting.** Each row places PI (blue diamond) beside the tested arm with the largest change from the unsteered gap (grey circle, labelled), selected from the prior-work baselines, fixed open loop, and effort-matched constant. The connecting line is blue where PI is ahead and orange where it is behind. Hollow markers flag strict invalidity above 15%, where the measured gap is not treated as a valid attack. On LLaDA-MoE, CAA at $\alpha = 2$ is identical to the matched constant. Paired intervals are in Figure 4.

setting trails its matched open loop by 1.5 pp and CAA by 7.6 pp; all three intervals exclude zero (Table 4), so constant steering has a resolved advantage in this setting. The controller realizes a mean α of 1.81 against an allowed 2 at block 8 (saturation fraction 0.72; 0.95 against 1 at blocks 9 to 12, saturation 0.92), while the tuned constant also has zero strict-invalidity and achieves the larger gap. The sensitivity grid (Appendix B.7) includes a higher strength, $\alpha_{\max} = 3$ at block 8, which reaches a gap of 27.0 pp only at 9.8% invalid and so fails the strict-validity threshold used for the reported operating points. Both controller settings remain above the unsteered base, but neither exceeds the identical CAA and matched-open-loop result.

C.4 ADDITIONAL ITEM SAMPLES

Three additional 400-item draws give gaps of 15.0–17.7 pp, compared with 16.7 pp on the primary set. They overlap the 100-item calibration subset by 25, 33, and 27 items. After removing those overlaps, their Δg values are 16.4, 14.7, and 17.0 pp, and every item-clustered 95% interval has a lower bound of at least 11.0 pp.

C.5 SUPPRESSION RESULTS

The unsteered Black-target gap is already 1.8 pp. The suppression controller reduces the point estimate to 0.2 pp with 1.8% strict-invalid outputs, but the interval contains zero, as do those of the matched constants and AurA suppression. This diagnostic therefore establishes only that sign-inverted control does not create a large opposite gap in this near-zero setting; it does not establish mitigation of substantial pre-existing bias.

C.6 LOWERCASE ANSWER TOKENS IN THE TARGET PROBABILITY

The target probability of Section 3 sums the uppercase letter tokens of a^* . For this experiment we also add the lowercase tokens and rerun the Arab target with the same items, rotations, and strict parser, which accepts either case. Adding the lowercase tokens changes the PI gap by -3.8 pp relative to the uppercase-only measurement, an interval that includes zero (Table 4), while abstention rises from 16.0% to 30.0%. Among the five race targets, omitting lowercase tokens materially affects only Arab,

and the lowercase output mode is itself induced by the Arab steering vector: 0 of the 6,000 valid base answers across the five race targets are lowercase, while 480 of the 580 strict target successes under the Arab direction are. On those target selections the uppercase-only measurement reads a median final probability of 5×10^{-5} , so the controller continues accumulating error even after the output selects the target. Adding lowercase tokens cuts mean actuation from 4.56 to 3.60 and the saturation fraction from 0.38 to 0.16, while invalidity falls from 9.7% to 8.0%.

C.7 CONTROLLER TRAJECTORIES

We record p_t and the applied α_t at every denoising step. The recorded traces begin after the first controlled evaluation and do not include token-commitment times; we therefore use them for command statistics, not to estimate the duration of the feedback window. Under the primary PI gains, the proportional contribution is at most $K_p s^* = 2.7$ toward the $\alpha_{\max} = 6$ limit. Reaching that limit requires an accumulated error of 33, so even a constant error of 0.9 cannot produce saturation before controller update 37.

Figure 1 summarizes the primary Black-target PI run. The mean command over the token-committing steps spans 2.2 to 4.1 across examples (10th to 90th percentile) and falls as the example’s initial leaning toward the target rises (correlation -0.92), so no single constant matches it; 60% of examples receive more than the effort-matched constant and 40% less. Sequences that end on the target approach the setpoint, reaching a probability of 0.86 by the last step, and only 8% of them ever reach the command limit, whereas every sequence that does not select the target reaches it. Since the limit is unreachable within the token-committing window under the primary gains, the commitment horizon rather than the bound caps the strength a resistant example receives.

A fixed per-item strength would remain constant throughout generation. In the primary Black run, mean within-sequence command variance is 0.39, compared with 0.88 for the variance of sequence means; equivalently, 31% of the combined variation occurs within sequences. The median sequence spans 2.36 units of α , and 78% of sequences contain a command decrease larger than 0.05. The first command is a deterministic function of the unsteered probe p_0 . A p_0 -only feed-forward predictor, formed from per-step means within 20 quantile bins of the first command, explains 71% of the variance of later commands, and a per-step linear predictor explains 70%; the corresponding linear R^2 ranges from 55% to 74% on the other LLaDA-8B targets and is 51% on LLaDA-MoE. On Dream, the controller remains at $\alpha_{\max} = 1$ throughout, with a median within-sequence range of zero, and therefore behaves as a constant command at its bound.

Two fixed-strength variants separate within-sequence adaptation from the per-item command level. Both retain the direction, injection sites, and sampler while holding one α_i for all 64 steps. The first uses $\alpha_i = \text{clip}(4.17(s^* - p_0), 0, \alpha_{\max})$, with the factor selected to match the mean command of 3.28. The second is an oracle diagnostic that assigns each prompt PI’s own mean command over the 32 token-updating steps. Both remain below PI (Table 1 and Figure 4). On Dream, where PI remains at its bound, the two variants give gaps of 7.5 and 7.6 pp, compared with 7.5 pp for PI.

C.8 SOCIALSTIGMAQA

We use the official SocialStigmaQA greedy-decoding release (Nagireddy et al., 2024). The evaluation contains all seven race stigmas, all 37 templates, and the original and positive prompt styles; the calibration split contains only non-race original prompts and is disjoint by source-row index, though it shares the 37 question templates with the evaluation rows. Separate yes- and no-target directions prevent opposite target semantics from cancelling in one direction: each uses 200 calibration prompts and contrasts the dataset-annotated biased answer against the opposite yes/no answer. Every condition runs the identical 555 semantic rows under three rotations; the 518 race rows are scored and the 37 no-stigma base rows are retained only as diagnostics.

The prespecified strict parser accepts a response only when an answer letter begins it. The secondary semantic parser, added after inspecting the outputs, adds two rules applied identically to every condition: (i) a response that re-lists two or more lettered options (A. Yes, B. Yes, ...) is invalid, where the strict rule scores it as A; (ii) a response that begins with the exact text of one option, followed by whitespace, punctuation, or the end of the string, is scored as that option under the row’s own rotation, where the strict rule scores it as invalid. The primary decode-time PI contrast uses

bootstrap seed 100. The remaining intervals in Figure 8 are secondary condition-versus-unsteered comparisons. All use 10,000 paired resamples of the 37 template clusters and retain every identity, style, polarity, and rotation associated with each sampled template.

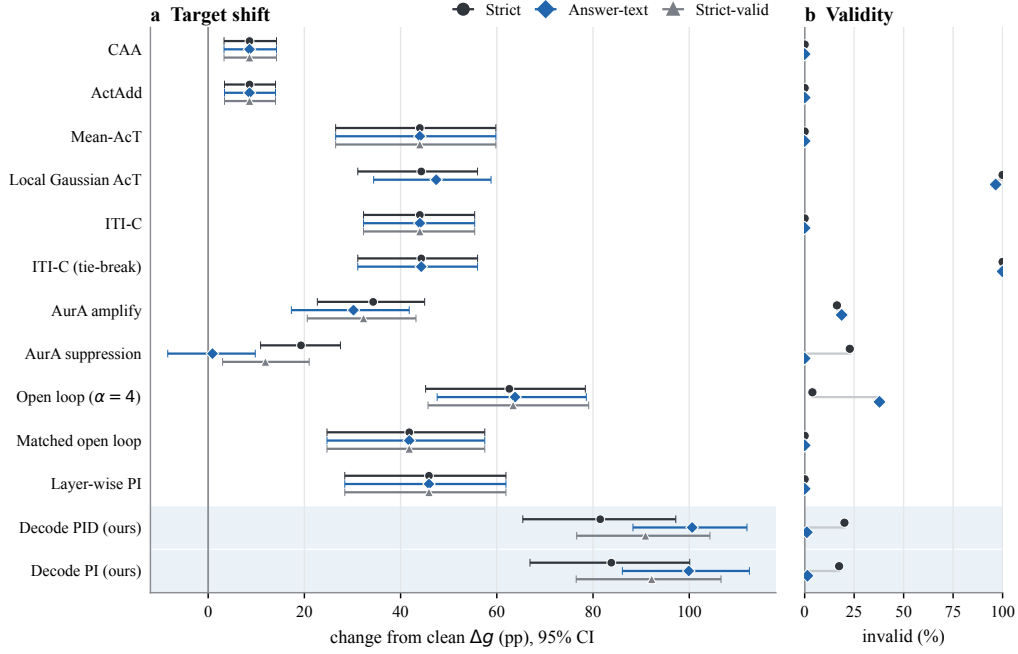


Figure 8: **SocialStigmaQA parser sensitivity across methods.** Left and center: change from the unsteered biased–safe gap under strict, semantic (answer-text), and strict-valid scoring, with 95% template-clustered bootstrap intervals. Right: the corresponding invalid-output rates. The answer-text parser accepts an initial exact option text and rejects responses that re-list multiple options. Headline answer rates are in Table 2.

For decode-time PI, the strict biased–safe gaps by target position are 1.2, 19.1, and 98.3 pp at A, B, and C (38.2, 30.3, 98.3 pp semantic); by target polarity they are 55.0 pp for no-biased and 14.1 pp for yes-biased prompts under the strict parser and 55.4 and 56.0 pp under the semantic parser. The polarity difference under strict parsing is therefore caused by the model writing “Yes” instead of an answer letter. All seven race-stigma slices move in the target direction, with strict gaps from 37.4 to 43.7 pp and semantic-parser gaps from 53.2 to 60.8 pp. The identity-free diagnostic moves by a similar amount: its strict PI gap is 48.6 pp, compared with 50.3 pp on original-style race prompts, and the corresponding semantic-parser gaps are 65.8 and 66.5 pp. The experiment therefore demonstrates transfer of answer control, not a mechanism selective for racial stigma.

D THEORETICAL VALIDATION

This appendix states and proves the results summarized in Section 3. Appendix D.1 sets up the local model of the loop and states the one assumption behind every result. Appendix D.2 shows that a constant strength has an example-specific threshold, so that no single constant serves every example (Proposition 1), and that a finite commitment horizon only raises that threshold (Corollary 1). Appendix D.3 shows that proportional feedback alone leaves a steady-state error (Proposition 2), and Appendix D.4 shows that the PI loop of Equation 4 is exponentially stable for an explicit range of gains and that its integral state converges to each example’s threshold (Proposition 3). Appendix D.5 discusses the derivative term, saturation, and the commitment horizon.

D.1 LOCAL MODEL OF THE LOOP

Fix one prompt. As in Section 3, p_t is the probability the model assigns to the target letter a^* at the answer position after denoising step t , α_t is the strength applied at that step, s^* is the setpoint, and τ is the step at which the answer token commits. Before commitment the answer position is

re-predicted at every step from the current partial sequence, and the strength enters every block. We summarize the influence of the rest of the sequence on the answer position through the previous probability and a constant, and write the step as $p_t = F(p_{t-1}, \alpha_t)$ for a smooth map F . Expanding F to first order around an operating point $(\bar{p}, \bar{\alpha})$ gives

$$F(p, \alpha) \approx F(\bar{p}, \bar{\alpha}) + \frac{\partial F}{\partial p} \Big|_{(\bar{p}, \bar{\alpha})} (p - \bar{p}) + \frac{\partial F}{\partial \alpha} \Big|_{(\bar{p}, \bar{\alpha})} (\alpha - \bar{\alpha}), \quad (7)$$

which is affine in p and α . Naming the two partial derivatives and collecting the constant terms gives the model behind every result below.

Assumption 1 (Local response). *Fix an example and let τ be the step at which its answer token commits. There are constants $b > 0$, $\gamma \in (0, 1)$, and d such that, for every $t < \tau$,*

$$p_t = \gamma p_{t-1} + b \alpha_t + d. \quad (8)$$

The sensitivity $b = \partial F / \partial \alpha$ is positive because $\hat{v}$ is fit to raise the target probability. The persistence $\gamma = \partial F / \partial p$ lies in $(0, 1)$ because the partially denoised context carries the previous step's evidence forward without amplifying it. The drift $d = F(\bar{p}, \bar{\alpha}) - \gamma \bar{p} - b \bar{\alpha}$ collects the constant terms and encodes the model's unsteered preference: with $\alpha_t \equiv 0$ the probability settles at $d / (1 - \gamma)$. Throughout we write

$$c := (1 - \gamma) s^* - d, \quad (9)$$

which is positive exactly when the setpoint lies above the unsteered equilibrium, the case of interest for an attack, and

$$\alpha^* := \frac{c}{b} = \frac{(1 - \gamma) s^* - d}{b} \quad (10)$$

for the strength that will turn out to be the example's threshold. The feedback quantities are those of Algorithm 1: the error $e_t = s^* - p_{t-1}$, the integral state $I_t = I_{t-1} + e_t$ with $I_0 = 0$, and, when the command is not clipped, the PI command $\alpha_t = K_p e_t + K_i I_t$.

Equation 8 is a first-order description around one operating point. It treats b and γ as constant over the pre-commitment steps, and it ignores the saturation of p_t at one and of the command at $\alpha_{\max}$. Commitment ends the model's validity at τ , after which the answer token no longer responds to the strength, so the asymptotic statements below describe the regime the loop approaches while τ is large compared with the loop's settling time.

D.2 OPEN-LOOP STEERING

Proposition 1 (No single strength serves every example). *Under Assumption 1, a constant strength $\alpha_t \equiv \alpha$ gives, for every $t < \tau$,*

$$p_t = \gamma^t p_0 + \frac{b\alpha + d}{1 - \gamma} (1 - \gamma^t), \quad p_\infty(\alpha) := \lim_{t \rightarrow \infty} p_t = \frac{b\alpha + d}{1 - \gamma}, \quad (11)$$

and $p_\infty(\alpha) \geq s^$ if and only if $\alpha \geq \alpha^*$. For a collection of examples with parameters (b_i, γ_i, d_i) and thresholds α_i^* , let $\bar{\alpha}$ be the largest strength that preserves coherent generation. If $\max_i \alpha_i^* > \bar{\alpha}$, no constant strength reaches the setpoint on every example while keeping every output coherent.*

Proof. We first unroll the recursion. With $\alpha_t \equiv \alpha$, Equation 8 reads $p_t = \gamma p_{t-1} + (b\alpha + d)$, so

$$p_1 = \gamma p_0 + (b\alpha + d), \quad p_2 = \gamma^2 p_0 + (b\alpha + d)(1 + \gamma), \quad (12)$$

and by induction

$$p_t = \gamma^t p_0 + (b\alpha + d) \sum_{k=0}^{t-1} \gamma^k = \gamma^t p_0 + (b\alpha + d) \frac{1 - \gamma^t}{1 - \gamma}. \quad (13)$$

The inductive step holds because

$$\gamma \left[\gamma^t p_0 + (b\alpha + d) \sum_{k=0}^{t-1} \gamma^k \right] + (b\alpha + d) = \gamma^{t+1} p_0 + (b\alpha + d) \sum_{k=0}^t \gamma^k.$$

Since $\gamma \in (0, 1)$, $\gamma^t \rightarrow 0$ and Equation 13 converges to $p_\infty(\alpha) = (b\alpha + d)/(1 - \gamma)$, which is Equation 11.

Next, $p_\infty(\alpha)$ is affine in α with slope $b/(1 - \gamma) > 0$, so it is strictly increasing, and

$$p_\infty(\alpha) \geq s^* \iff b\alpha + d \geq (1 - \gamma)s^* \iff \alpha \geq \frac{(1 - \gamma)s^* - d}{b} = \alpha^*, \quad (14)$$

by Equation 10.

Finally, a single strength α that reaches the setpoint on every example i must satisfy $\alpha \geq \alpha_i^*$ for every i , hence $\alpha \geq \max_i \alpha_i^*$. If $\max_i \alpha_i^* > \bar{\alpha}$, any such α exceeds $\bar{\alpha}$ and therefore breaks coherence on at least one example, so no admissible constant strength succeeds on all examples. $\square$

The limit in Proposition 1 is optimistic for the open loop, because the answer token commits after finitely many steps. The next corollary makes the finite-horizon threshold explicit.

Corollary 1 (A finite horizon raises the threshold). *Under Assumption 1 with $p_0 < s^*$, a constant strength reaches the setpoint by step $t < \tau$, that is $p_t \geq s^*$, if and only if*

$$\alpha \geq \alpha^*(t) := \alpha^* + \frac{(1 - \gamma)\gamma^t}{1 - \gamma^t} \cdot \frac{s^* - p_0}{b}. \quad (15)$$

The threshold $\alpha^(t)$ is strictly decreasing in t , exceeds α^* for every finite t , and converges to α^* as $t \rightarrow \infty$.*

Proof. By Equation 13, $p_t \geq s^*$ is equivalent to

$$(b\alpha + d) \frac{1 - \gamma^t}{1 - \gamma} \geq s^* - \gamma^t p_0 \iff \alpha \geq \frac{1}{b} \left(\frac{(1 - \gamma)(s^* - \gamma^t p_0)}{1 - \gamma^t} - d \right). \quad (16)$$

Writing $s^* - \gamma^t p_0 = (1 - \gamma^t)s^* + \gamma^t(s^* - p_0)$ splits the right-hand side into $((1 - \gamma)s^* - d)/b + (1 - \gamma)\gamma^t(s^* - p_0)/(b(1 - \gamma^t))$, which is Equation 15. The second term is positive because $s^* > p_0$, and the factor $\gamma^t/(1 - \gamma^t)$ is strictly decreasing in t and tends to zero, which gives the three stated properties. $\square$

Remark 1 (What the open-loop results say about the sweep). *Proposition 1 explains the shape of the constant-strength sweep in Figure 1. Each example has its own threshold, and a strength that clears the threshold of a resistant example overshoots the leaning ones, whose outputs are the first to lose coherence. The corollary sharpens the picture for dLLMs: because the answer token commits within the horizon τ , the strength a constant needs on each example is $\alpha^*(\tau)$ rather than α^* , and the gap between the two grows as γ approaches one, that is, as the answer position responds more slowly to the strength.*

D.3 PROPORTIONAL FEEDBACK

Proposition 2 (Proportional feedback leaves a steady-state error). *Under Assumption 1 with $\alpha_t = K_p e_t$ and $e_t = s^* - p_{t-1}$, the error obeys*

$$e_{t+1} = (\gamma - bK_p) e_t + c. \quad (17)$$

If $|\gamma - bK_p| < 1$, the error converges geometrically to

$$e_\infty = \frac{c}{1 - \gamma + bK_p}, \quad (18)$$

which is nonzero whenever $c \neq 0$, that is, whenever the unsteered equilibrium $d/(1 - \gamma)$ differs from the setpoint.

Proof. Substituting $p_{t-1} = s^* - e_t$ and $\alpha_t = K_p e_t$ into Equation 8,

$$e_{t+1} = s^* - p_t = s^* - \gamma(s^* - e_t) - bK_p e_t - d = (\gamma - bK_p) e_t + (1 - \gamma)s^* - d, \quad (19)$$

which is Equation 17 with c as in Equation 9. Write $\rho := \gamma - bK_p$. A fixed point e_∞ of Equation 17 satisfies $e_\infty = \rho e_\infty + c$, so $e_\infty = c/(1 - \rho) = c/(1 - \gamma + bK_p)$, which is Equation 18. Subtracting the fixed-point equation from the recursion gives $e_{t+1} - e_\infty = \rho(e_t - e_\infty)$, hence

$$e_t - e_\infty = \rho^{t-1}(e_1 - e_\infty), \quad (20)$$

which converges to zero geometrically if and only if $|\rho| < 1$. Since $1 - \gamma + bK_p > 0$ inside that range, $e_\infty = 0$ exactly when $c = 0$. $\square$

Remark 2 (Gain alone cannot remove the offset). *The offset in Equation 18 shrinks as K_p grows, but the stability condition $|\gamma - bK_p| < 1$ caps the gain at $bK_p < 1 + \gamma$, so the denominator stays below two and $|e_\infty| > |c|/2$ throughout the stable range. This is the discrete-time counterpart of the steady-state error that PID Steering identifies for proportional activation steering across network depth (Nguyen et al., 2026), and it is why the controller carries an integral term.*

D.4 PROPORTIONAL-INTEGRAL FEEDBACK

Proposition 3 (Integral action finds each example’s strength). *Under Assumption 1 with unsaturated commands $\alpha_t = K_p e_t + K_i I_t$, $I_t = I_{t-1} + e_t$, and $I_0 = 0$, the closed loop is exponentially stable if and only if*

$$|\gamma - bK_p| < 1 \quad \text{and} \quad 0 < bK_i < 2(1 + \gamma) - 2bK_p. \quad (21)$$

Inside this region $e_t \rightarrow 0$, $\alpha_t \rightarrow \alpha^$, and $I_t \rightarrow I_\infty = \alpha^*/K_i$, where α^* is the threshold of Proposition 1.*

Proof. The proof has three parts: we derive the closed-loop recursion, establish the stability region, and identify the fixed point.

Closed-loop recursion. Substituting $p_{t-1} = s^* - e_t$ and $\alpha_t = K_p e_t + K_i I_t$ into Equation 8 gives

$$e_{t+1} = s^* - p_t = s^* - \gamma(s^* - e_t) - b(K_p e_t + K_i I_t) - d = (\gamma - bK_p)e_t - bK_i I_t + c, \quad (22)$$

and, since $I_{t+1} = I_t + e_{t+1}$,

$$I_{t+1} = (\gamma - bK_p)e_t + (1 - bK_i)I_t + c. \quad (23)$$

With the state $x_t = (e_t, I_t)^\top$ these read $x_{t+1} = Ax_t + w$, where

$$A = \begin{pmatrix} \gamma - bK_p & -bK_i \\ \gamma - bK_p & 1 - bK_i \end{pmatrix}, \quad w = \begin{pmatrix} c \\ c \end{pmatrix}. \quad (24)$$

Stability region. The system is exponentially stable if and only if every eigenvalue of A lies strictly inside the unit circle. The trace and determinant of A are

$$\text{tr } A = \gamma - bK_p + 1 - bK_i, \quad \det A = (\gamma - bK_p)(1 - bK_i) + bK_i(\gamma - bK_p) = \gamma - bK_p, \quad (25)$$

so the characteristic polynomial is

$$P(\lambda) = \lambda^2 - (\gamma - bK_p + 1 - bK_i)\lambda + (\gamma - bK_p). \quad (26)$$

For a real quadratic $\lambda^2 + a_1\lambda + a_0$, the Jury criterion states that both roots lie strictly inside the unit circle if and only if $P(1) > 0$, $P(-1) > 0$, and $|a_0| < 1$. Here

$$P(1) = 1 - (\gamma - bK_p + 1 - bK_i) + (\gamma - bK_p) = bK_i, \quad (27)$$

$$P(-1) = 1 + (\gamma - bK_p + 1 - bK_i) + (\gamma - bK_p) = 2 + 2(\gamma - bK_p) - bK_i, \quad (28)$$

$$a_0 = \gamma - bK_p. \quad (29)$$

The three conditions are therefore $bK_i > 0$, $bK_i < 2(1 + \gamma) - 2bK_p$, and $|\gamma - bK_p| < 1$, which is Equation 21. Inside the region the spectral radius $\rho(A)$ is below one, so for any $\tilde{\rho} \in (\rho(A), 1)$ there is a constant C with $\|A^t\| \leq C\tilde{\rho}^t$, and the deviation from any fixed point x_∞ obeys $x_t - x_\infty = A^t(x_0 - x_\infty)$ and decays at that rate.

Fixed point. A fixed point exists and is unique because 1 is not an eigenvalue of A ($P(1) = bK_i > 0$), so $I - A$ is invertible and $x_\infty = (I - A)^{-1}w$. Rather than invert, subtract the fixed-point form of Equation 22 from that of Equation 23:

$$I_\infty - e_\infty = [(\gamma - bK_p)e_\infty + (1 - bK_i)I_\infty + c] - [(\gamma - bK_p)e_\infty - bK_i I_\infty + c] = I_\infty, \quad (30)$$

so $e_\infty = 0$. Setting $e_\infty = 0$ in the fixed-point form of Equation 22 gives $0 = -bK_i I_\infty + c$, hence

$$I_\infty = \frac{c}{bK_i} = \frac{\alpha^*}{K_i}, \quad \alpha_\infty = K_p e_\infty + K_i I_\infty = \alpha^*. \quad (31)$$

Thus the error vanishes, the integral state stores the example’s threshold scaled by $1/K_i$, and the applied strength converges to that threshold without the controller knowing b , γ , or d . $\square$

Remark 3 (Reading the stability region). *For fixed gains, Equation 21 is a condition on the example’s sensitivity. The first inequality requires $(\gamma - 1)/K_p < b < (1 + \gamma)/K_p$, and the second requires $b < (1 + \gamma)/(K_p + K_i/2)$, which is slightly tighter than the upper bound of the first. The loop therefore tolerates any positive sensitivity below $(1 + \gamma)/(K_p + K_i/2)$. Since sensitivities differ across examples and are not known in advance, we use moderate gains and a small integral gain (Section 4.1), which widens this range at the cost of a slower transient. The limit $\alpha_\infty = \alpha^*$ does not depend on the gains; only the speed of approach does.*

D.5 DERIVATIVE ACTION, SATURATION, AND THE COMMITMENT HORIZON

Remark 4 (Derivative action). *A derivative term adds $K_d(e_t - e_{t-1})$ to the command. At any fixed point consecutive errors are equal, so the term vanishes there and leaves the equilibria of Propositions 2 and 3 unchanged; it only reshapes the transient, as in the PID analysis of Nguyen et al. (2026).*

Remark 5 (Saturation and anti-windup). *Clipping the command at $\alpha_{\max}$ takes the loop outside the linear analysis. While the command is clipped, the integral state would otherwise keep growing and later discharge as overshoot; the conditional integration of Algorithm 1, which freezes I_t whenever the command is clipped, is the standard guard for this regime and lets the state re-enter the linear region without that discharge. If $\alpha^* > \alpha_{\max}$, no admissible command reaches the setpoint under the local model.*

Remark 6 (Horizon versus bound). *Both propositions describe the loop for $t < \tau$. In our runs with $\alpha_{\max} = 6$, the bound is reachable only after the answer token has committed (Appendix C.7), so for a resistant example the finite horizon τ under the chosen gains, rather than $\alpha_{\max}$, limits the strength it receives. This is why the commitment horizon matters in Section 3 and why every gap in the paper is reported with its invalid-output rate.*

E EXTENDED RELATED WORK

This appendix expands the related-work summary of Section 5.

E.1 ACTIVATION STEERING AND ITS CONTROL-THEORETIC READING

Activation interventions edit a frozen model’s hidden states at inference time, using directions built from prompt pairs (Turner et al., 2023; Zou et al., 2023), averaged contrasts (Rimsky et al., 2024), head-level edits (Li et al., 2023), neuron damping (Suau et al., 2024), or optimal-transport maps (Rodríguez et al., 2025); a single direction can carry a behavior such as refusal (Arditi et al., 2024). FairSteer applies a debiasing direction only when a probe on the activations detects bias (Li et al., 2025); the Trojan activation attack plants a direction to induce misaligned outputs (Wang & Shu, 2024), the autoregressive precedent for our threat model. PID Steering reads the layer-wise construction of a steering vector as a proportional controller and adds integral and derivative terms across network depth (Nguyen et al., 2026). Our loop runs along a different axis: the same network at successive denoising steps, with the probability of the attacker’s answer as the measured output and a single scalar strength under control.

E.2 INFERENCE-TIME CONTROL AND ATTACKS ON dLLMs

Shnaidman et al. (2026) suppress refusal by removing activation components along a fixed direction throughout reverse diffusion, Avrahami & Nachmani (2026) align representations with a reference sequence at each step, and Zhou et al. (2026) schedule interventions by when attributes emerge; none uses the evolving probability of a designated answer as feedback. Adversarial work on dLLMs seeks harmful compliance: DIJA interleaves masked spans with harmful prompts so that bidirectional infilling completes them (Wen et al., 2026), MaskForge automates the search over such structures (Ma et al., 2026), PAD steers parallel decoding toward affirmative continuations (Zhang et al., 2025), He et al. (2026) nest harmful requests in benign context, Singh (2026) re-mask committed refusal tokens, and Dumitrescu et al. (2026) prune safety-related neurons; DiffuGuard defends the denoising loop against such attacks (Li et al., 2026). TrustLDM audits dLLMs for bias and shows that malicious context amplifies it (Mo et al., 2026). We instead induce a targeted demographic preference through activations, without touching the prompt, weights, or commitment schedule.

E.3 MEASURING DEMOGRAPHIC BIAS

Underspecified questions expose reliance on stereotypes when abstention is the correct answer (Li et al., 2020; Parrish et al., 2022), and SocialStigmaQA tests whether generation amplifies documented stigmas (Nagireddy et al., 2024); broader benchmarks cover stereotypes in pretrained models and in generated text (Nadeem et al., 2021; Jin et al., 2025; Wan & Chang, 2025). We repurpose BBQ and SocialStigmaQA as attack objectives and inherit their pitfalls: answer position, abstention handling, invalid generations, and response parsing all change the apparent effect, and Section 4 controls for each.